Preemptive Clinical Pharmacogenomics Implementation: A Comprehensive Literature Review of Study Outcomes

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Background

- Pharmacogenomics (PGx) is used to:
  - Identify patients at risk of having adverse drug reactions based on genetics
  - Guide drug dosing and selection to optimize patient outcomes
- Preemptive PGx informs initial prescribing decisions
- Outcomes characterize clinical effectiveness of PGx-informed treatment, but vary greatly between studies
- The objective of this work was to conduct a comprehensive literature review of study outcomes used in preemptive clinical PGx implementation studies

Methods

- Study selection is shown as a flowchart in Figure 1
- Inclusion Criteria:
  - Preemptive clinical PGx implementation study (prospective, retrospective, sub-study, or protocol) with PGx-informed patient care and outcomes
- Exclusion Criteria:
  - Basic research, guidelines, reviews, commentary, cost, clinical decision support, and surveys

Figure 1: Study Selection Flowchart

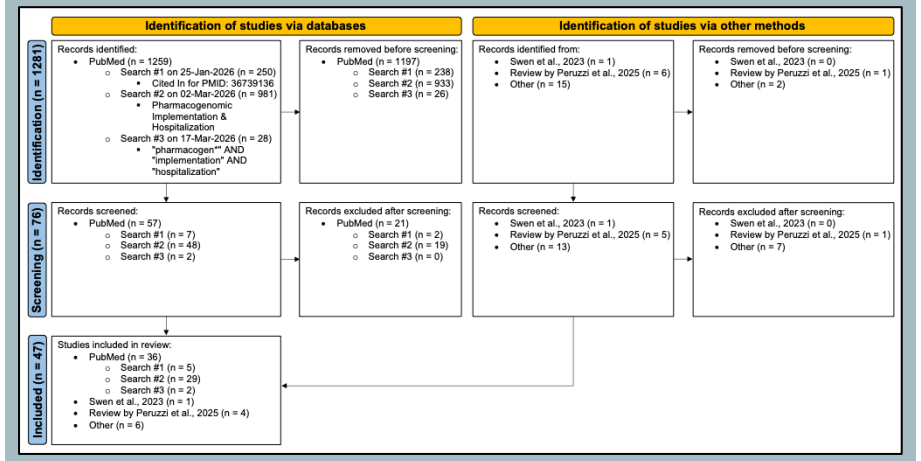
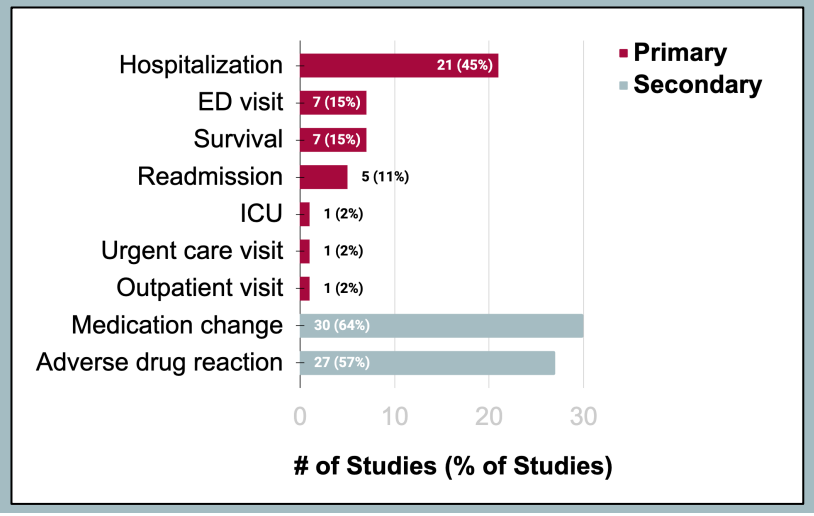


Figure 2: Summary of Outcomes from 47 Studies



Conclusions

- Study outcomes are summarized in Figure 2
- Most common:
  - Primary outcome: Hospitalization (45%)
  - Secondary outcome: Medication change (64%)
- Least common:
  - Primary outcomes: ICU, Urgent care visit, and Outpatient visit (2% each)

Future Directions

- Future work will evaluate primary (e.g., hospitalization) and secondary (e.g., medication changes) outcomes following preemptive clinical PGx implementation in Colorado Center for Personalized Medicine biobank participants

Select References

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- Page et al., 2021. doi: 10.1136/bmj.n71